

Practice: Build Bus Schedule app

1. Before you begin

Introduction

In the [Persist Data with Room](#) codelab, you learned how to implement a Room database in an Android app. This exercise provides the opportunity to gain more familiarity with the implementation of Room databases through an independently driven set of steps.

In this practice set, you take the concepts you learned from the [Persist Data with Room](#) codelab to complete the Bus Schedule app. This app presents the user with a list of bus stops and scheduled departures using data provided from a Room database.

The solution code is available at the end. To make the most of this learning experience, try to implement and troubleshoot as much as you can before you look at the provided solution code. It is during this hands-on time that you learn the most.

Prerequisites

- Android Basics with Compose coursework through the [Persist Data with Room](#) codelab

What you'll need

- A computer with internet access and Android Studio
- The Bus Schedule starter code

What you'll build

In this practice set, you complete the Bus Schedule app by implementing a database and then delivering data to the UI using the database. A database file in the asset directory found in the starter code provides data for the app. You load this data into a database and make it available for read usage by the app.

After you complete the app, it shows a list of bus stops and corresponding arrival times. You can click an item in the list to trigger navigation to a detail screen that provides data for that stop.

The completed app shows this data, loaded from a Room database:

12:00



Bus Schedule

Stop Name

Arrival Time

Main Street

11:00 AM

Park Street

11:12 AM

Maple Avenue

11:25 AM

Broadway Avenue

11:41 AM

Post Street

11:58 AM

Elm Street

12:09 PM

Oak Drive

12:20 PM

Middle Street

12:34 PM

Palm Avenue

12:51 PM

Winding Way

12:55 PM

Main Street

1:00 PM

Park Street

1:12 PM

Maple Avenue

1:25 PM

12:00



← Post Street

Post Street Stop

Arrival Time

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11:58 AM

--

1:58 PM

--

3:58 PM

2. Download the starter code

Starter code URL:

<https://github.com/google-developer-training/basic-android-kotlin-compose-training-bus-schedule-app>

Branch name with starter code: starter

1. In Android Studio, open the basic-android-kotlin-compose-training-bus-schedule folder.
2. Open the Bus Schedule app code in Android Studio.



3. Click the **Run** button to build and run the app.

The app is expected to display a schedule showing one stop when built from the starter branch code.

12:00



Bus Schedule

Stop Name

Arrival Time

Example Street

7:00 PM

3. Add dependencies

Add the following dependencies to the app:

app/build.gradle.kts

```
implementation("androidx.room:room-ktx:${rootProject.extra["room_version"]}")
implementation("androidx.room:room-runtime:${rootProject.extra["room_version"]}")
ksp("androidx.room:room-compiler:${rootProject.extra["room_version"]}")
```

You should get the most current stable version of room from the [Room documentation](#) and add the correct version number. At this moment the latest version is:

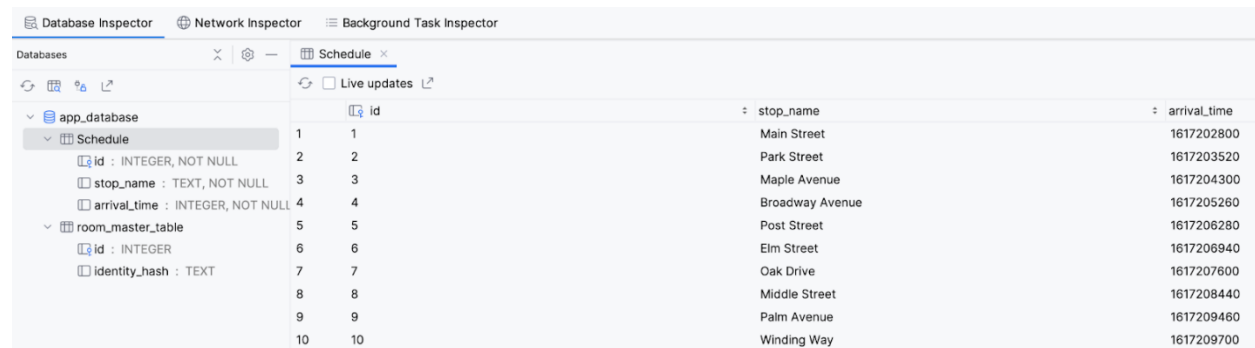
build.gradle.kts

```
set("room_version", "2.5.1")
```

4. Create a Room entity

Convert the current Bus Schedule data class into a Room Entity.

The following image shows a sample of what the final data table looks like, including the schema and Entity property.



The screenshot shows the Database Inspector in Android Studio. The left pane shows the database structure: app_database contains a Schedule table (id: INTEGER, NOT NULL; stop_name: TEXT, NOT NULL; arrival_time: INTEGER, NOT NULL) and a room_master_table (id: INTEGER; identity_hash: TEXT). The right pane shows the data for the Schedule table, which is ordered by arrival_time. The data includes 10 rows of bus stops and their arrival times.

id	stop_name	arrival_time
1	Main Street	1617202800
2	Park Street	1617203520
3	Maple Avenue	1617204300
4	Broadway Avenue	1617205260
5	Post Street	1617206280
6	Elm Street	1617206940
7	Oak Drive	1617207600
8	Middle Street	1617208440
9	Palm Avenue	1617209460
10	Winding Way	1617209700

5. Create a data access object

Create a data access object (DAO) to access the database. The DAO provides a method to retrieve all the items in the database and a method to retrieve a single item with the name of the bus stop. Make sure to order the schedule by arrival time.

6. Create a database instance

Create a Room database that uses the Entity and your DAO. The database initializes itself with data from the assets/database/bus_schedule.db file in the starter code.

7. Update the ViewModel

Update the ViewModel to retrieve data from the DAO and provide it to the UI instead of supplying sample data. Make sure to leverage both of your DAO methods to supply data for the list and for individual stops.

8. Solution code

Solution code URL:

<https://github.com/google-developer-training/basic-android-kotlin-compose-training-bus-schedule-app>

Branch name with solution code: main